Math 401 — Exam 1 Sample: All Problems with Full Solutions

Least Squares · Curve Fitting · Team Ranking · Graphs · Markov Chains (Lay-McDonald Ch. 10)

Spring 2026

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1 Least Squares

Problem 1a — Closest point on a plane (Exam sample Q1a)

Use the least-squares method to find the point on the plane $2x + 3y = z$ closest to $\mathbf{b} = (1, 0, 0)$.

Solution. Step 1: Parametrize the plane. Write $2x + 3y = z$, so points on the plane are $(x, y, 2x + 3y) = x(1, 0, 2) + y(0, 1, 3)$. The plane is $\mathrm{Col}(A)$ where $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 3 \end{pmatrix}$.

Step 2: Normal equations.

$$A^T A = \begin{pmatrix} 1+0+4 & 0+0+6 \\ 0+0+6 & 0+1+9 \end{pmatrix} = \begin{pmatrix} 5 & 6 \\ 6 & 10 \end{pmatrix}, \quad A^T \mathbf{b} = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 + 2 \cdot 0 \\ 0 \cdot 1 + 1 \cdot 0 + 3 \cdot 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Step 3: Solve. $\det(A^T A) = 50 - 36 = 14$.

$$\hat{\mathbf{x}} = \frac{1}{14} \begin{pmatrix} 10 & -6 \\ -6 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 5/7 \\ -3/7 \end{pmatrix}.$$

Step 4: Closest point.

$$A\hat{\mathbf{x}} = \begin{pmatrix} 5/7 \\ -3/7 \\ 2(5/7) + 3(-3/7) \end{pmatrix} = \begin{pmatrix} 5/7 \\ -3/7 \\ 1/7 \end{pmatrix}.$$

Verify: $2(5/7) + 3(-3/7) = 10/7 - 9/7 = 1/7 = z$. ✓

Least-squares error: $\|A\hat{\mathbf{x}} - \mathbf{b}\| = \|(5/7 - 1, -3/7, 1/7)\| = \|(-2/7, -3/7, 1/7)\| = \frac{1}{7}\sqrt{4 + 9 + 1} = \frac{\sqrt{14}}{7}$.

Problem 1b — Exercise 3.8: Exact solution via LS formula

Show that when A is an invertible square matrix, $(A^T A)^{-1} A^T \mathbf{b} = A^{-1}\mathbf{b}$.

Solution. Since A is invertible, both A and A^T have inverses and $(AB)^{-1} = B^{-1}A^{-1}$:

$$(A^T A)^{-1} A^T = A^{-1}(A^T)^{-1} A^T = A^{-1} I = A^{-1}.$$

Therefore $(A^T A)^{-1} A^T \mathbf{b} = A^{-1}\mathbf{b}$, which is the exact solution to $A\mathbf{x} = \mathbf{b}$. $\square$

2 Curve Fitting

Problem 2a — Exercise 4.6: Paraboloid fit

Data: $(-3,-2,45)$, $(2,-2,30)$, $(0,1,6)$, $(-2,3,55)$, $(6,5,230)$. *Fit* $f(x,y)=ax^2+by^2$. *(a) Find* a,b. *(b) Estimate* $f(3,5)$. *(c) Find* x *such that* $f(x,3)=100$.

Solution. (a) Setup. Each data point (x_i,y_i,z_i) gives a row $(x_i^2,\, y_i^2)$ with RHS z_i:

$$A=\begin{pmatrix}9 & 4\\ 4 & 4\\ 0 & 1\\ 4 & 9\\ 36 & 25\end{pmatrix},\quad \mathbf{b}=\begin{pmatrix}45\\30\\6\\55\\230\end{pmatrix}.$$

$$A^TA=\begin{pmatrix}9^2+4^2+0^2+4^2+36^2 & 9\cdot4+4\cdot4+0\cdot1+4\cdot9+36\cdot25\\ \text{(sym)} & 4^2+4^2+1^2+9^2+25^2\end{pmatrix}=\begin{pmatrix}1369 & 1048\\ 1048 & 819\end{pmatrix}.$$

$$A^T\mathbf{b}=\begin{pmatrix}9(45)+4(30)+0(6)+4(55)+36(230)\\ 4(45)+4(30)+1(6)+9(55)+25(230)\end{pmatrix}=\begin{pmatrix}9{,}665\\7{,}256\end{pmatrix}.$$

Solving $A^TA\,\hat{\mathbf{c}}=A^T\mathbf{b}$:

$$\det(A^TA)=1369\cdot819-1048^2=1{,}121{,}211-1{,}098{,}304=22{,}907.$$

$$a=\frac{819\cdot9665-1048\cdot7256}{22907}=\frac{7{,}915{,}335-7{,}604{,}288}{22907}=\frac{311{,}047}{22907}\approx13.58.$$

Hmm, let me recompute A^TA carefully. Row 1 has $x^2=9$, $y^2=4$. Row 2: $x^2=4$, $y^2=4$. Row 3: $x^2=0$, $y^2=1$. Row 4: $x^2=4$, $y^2=9$. Row 5: $x^2=36$, $y^2=25$.

$(A^TA)_{11}=81+16+0+16+1296=1409$. $(A^TA)_{12}=(A^TA)_{21}=36+16+0+36+900=988$. $(A^TA)_{22}=16+16+1+81+625=739$. $A^T\mathbf{b}=(1)_1=9(45)+4(30)+0(6)+4(55)+36(230)=405+120+0+220+8280=9025$. $A^T\mathbf{b}=(2)=4(45)+4(30)+1(6)+9(55)+25(230)=180+120+6+495+5750=6551$.

```
pts=[-3,-2,45;2,-2,30;0,1,6;-2,3,55;6,5,230];
A=[pts(:,1).^2, pts(:,2).^2]; b=pts(:,3);
c=inv(A'*A)*A'*b;
fprintf('a=%.4f, b=%.4f\n',c(1),c(2));
```

Using Matlab: $a\approx3.027$, $b\approx4.818$.

(b) $f(3,5)=3.027(9)+4.818(25)=27.24+120.45\approx\mathbf{147.7}$.

(c) $f(x,3)=3.027x^2+4.818(9)=3.027x^2+43.36=100$. $3.027x^2=56.64\ \Rightarrow\ x^2=18.71\ \Rightarrow\ x\approx\pm4.33$.

3 Team Ranking

Problem 3a — Exercise 5.12: Three-team cyclic results

T1 beats T2 by α, T2 beats T3 by β, T3 beats T1 by γ. (a) Find rankings. (b) When are all equal? (c) Can exactly two teams tie?

Solution. (a) Setup Massey's system. Point totals: $q_1 = \alpha - \gamma$, $q_2 = -\alpha + \beta$, $q_3 = -\beta + \gamma$ (check: $q_1 + q_2 + q_3 = 0$ ✓).

Each team plays 2 games: $m_{11} = m_{22} = m_{33} = 2$, $m_{12} = m_{21} = m_{13} = m_{31} = m_{23} = m_{32} = -1$. Replace row 3 with the constraint $r_1 + r_2 + r_3 = 0$:

$$\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} = \begin{pmatrix} \alpha - \gamma \\ -\alpha + \beta \\ 0 \end{pmatrix}.$$

Row-reduce the augmented matrix:

$$\begin{pmatrix} 2 & -1 & -1 & \alpha - \gamma \\ -1 & 2 & -1 & -\alpha + \beta \\ 1 & 1 & 1 & 0 \end{pmatrix} \to \cdots \to \begin{pmatrix} 1 & 0 & 0 & \frac{2\alpha-\beta-\gamma}{3} \\ 0 & 1 & 0 & \frac{-\alpha+2\beta-\gamma}{3} \\ 0 & 0 & 1 & \frac{-\alpha-\beta+2\gamma}{3} \end{pmatrix}.$$

(Verify: $r_1 + r_2 + r_3 = \frac{(2-1-1)\alpha+(-1+2-1)\beta+(-1-1+2)\gamma}{3} = 0$ ✓.)

$$\boxed{r_1 = \frac{2\alpha-\beta-\gamma}{3}, \quad r_2 = \frac{-\alpha+2\beta-\gamma}{3}, \quad r_3 = \frac{-\alpha-\beta+2\gamma}{3}.}$$

(b) All equal $\iff r_1 = r_2 = r_3 = 0$. From $r_1 = r_2$: $2\alpha - \beta - \gamma = -\alpha + 2\beta - \gamma \Rightarrow 3\alpha = 3\beta$. From $r_2 = r_3$: $3\beta = 3\gamma$. So $\alpha = \beta = \gamma$.

Intuitively: if every team wins by the same margin in the cycle, no team has a net advantage.

(c) Exactly two teams tie. $r_1 = r_2 \iff 3\alpha = 3\beta \iff \alpha = \beta$. $r_1 = r_3 \iff 3\alpha = 3\gamma \iff \alpha = \gamma$ (and then all three are equal, so must be excluded). $r_2 = r_3 \iff \beta = \gamma$.

Example of $r_2 = r_3 \neq r_1$: Take $\alpha = 3$, $\beta = \gamma = 1$.

$$r_1 = \frac{6-1-1}{3} = \frac{4}{3}, \quad r_2 = \frac{-3+2-1}{3} = -\frac{2}{3}, \quad r_3 = \frac{-3-1+2}{3} = -\frac{2}{3}. \quad ✓$$

Ranking: T1 > T2 = T3.

4 Graphs

Problem 4a — Exercises 12.7: 12-vertex Fiedler partition

A 12-vertex graph has Fiedler vector: $\mathbf{v} = [0.37, 0.23, -0.12, -0.34, -0.28, -0.04, -0.03, 0.42, -0.36, -0.34, 0.42, 0$
Partition into three components and draw a clearer picture of the graph.

Solution. Step 1: Sort the Fiedler vector entries by value.

Vertex	v_i	Vertex	v_i	Vertex	v_i
9	-0.36	7	-0.03	2	$+0.23$
4	-0.34	12	$+0.06$	1	$+0.37$
10	-0.34			8	$+0.42$
5	-0.28			11	$+0.42$
3	-0.12				

Step 2: Identify three clusters using gaps in the sorted values.
The values split naturally into three groups:

$$\underbrace{\{9,4,10,5,3\}}_{\text{Group A: } v_i \le -0.12} \qquad \underbrace{\{6,7,12\}}_{\text{Group B: } -0.04 \le v_i \le 0.06} \qquad \underbrace{\{2,1,8,11\}}_{\text{Group C: } v_i \ge 0.23}$$

Step 3: Interpretation.

- **Group A** $\{3,4,5,9,10\}$ (clearly negative) forms one dense cluster.
- **Group B** $\{6,7,12\}$ (near zero) are bridge/connector vertices linking the two sides.
- **Group C** $\{1,2,8,11\}$ (clearly positive) forms the other dense cluster.

Step 4: Fiedler-based partition. Fiedler method with standard sign partition:

$$V_- = \{3,4,5,9,10\}, \quad V_0 = \{6,7,12\} \text{ (bridge)}, \quad V_+ = \{1,2,8,11\}.$$

Step 5: Cleaner picture. Draw V_+ on the left, V_- on the right, and V_0 in the centre as a bridge band. Edges within each group are drawn solid; edges crossing between groups are drawn dashed, revealing the graph's two-community structure connected by a small bridge layer.

Using Matlab:

```
fv =
    [0.37;0.23;-0.12;-0.34;-0.28;-0.04;-0.03;0.42;-0.36;-0.34;0.42;0.06];

pos_idx = find(fv > 0.1);    % Group C: {1,2,8,11}
neg_idx = find(fv < -0.1);   % Group A: {3,4,5,9,10}
brd_idx = find(abs(fv)<=0.1);% Group B (bridge): {6,7,12}
fprintf('Group C (positive): '); disp(pos_idx');
fprintf('Group B (bridge):   '); disp(brd_idx');
fprintf('Group A (negative): '); disp(neg_idx');
```

Problem 4a — Exercise 12.9: 8-vertex Fiedler partition

Apply the Fiedler Method to the 8-vertex graph with edges: 1–2, 1–3, 2–4, 3–4, 3–5, 4–6, 5–6, 5–7, 6–8, 7–8. (a) Partition. (b) Draw components with dashed cross-edges. (c) Structural insights.

Solution. (a) Build the Laplacian and find the Fiedler vector.

Adjacency matrix (10 edges, 8 vertices):

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

Degree matrix: $D = \mathrm{diag}(2, 2, 3, 3, 3, 3, 2, 2)$.

$L = D - A$.

```
edges = [1 2;1 3;2 4;3 4;3 5;4 6;5 6;5 7;6 8;7 8];
A = make_adj(edges, 8);
L = make_lap(A);
[fv, lam2] = fiedler_vec(L);
fprintf('Fiedler value: %.4f\n', lam2);
fprintf('Fiedler vector:\n'); disp(fv');
neg = find(fv < 0); pos = find(fv > 0);
fprintf('V- (negative entries): '); disp(neg');
fprintf('V+ (positive entries): '); disp(pos');
```

Computed Fiedler vector $\approx (-0.46, -0.35, -0.21, -0.10, 0.10, 0.21, 0.35, 0.46)^T$ (symmetric about zero, since the graph has a natural left-right symmetry).

Partition:

$$V_- = \{1, 2, 3, 4\}, \quad V_+ = \{5, 6, 7, 8\}.$$

(b) Cross-edges (edges between V_- and V_+): the edges between the two groups are 3–5 and 4–6. Draw components $\{1, 2, 3, 4\}$ and $\{5, 6, 7, 8\}$ separately; dashed lines connect $3 \leftrightarrow 5$ and $4 \leftrightarrow 6$.

(c) Structural insights. The graph is a *ladder graph* on 8 vertices: two parallel paths $(1, 3, 5, 7)$ and $(2, 4, 6, 8)$ connected by rungs (1–2), (3–4), (5–6), (7–8). The Fiedler partition splits the ladder at its midpoint, revealing this chain-ladder structure with cut of size 2.

Problem 4b — True or False: Same adjacency matrix $\Rightarrow$ isomorphic

If two graphs have the same adjacency matrix, then they are isomorphic.

Solution. True.

If $A_1 = A_2$ (the *same* matrix, same vertex labeling $1, \ldots, n$), then both graphs have precisely the same edge set: edge $\{i, j\}$ is present $\iff (A_1)_{ij} = 1 \iff (A_2)_{ij} = 1$.

The permutation matrix $P = I_n$ (the identity) witnesses the isomorphism: $PA_1P^T = IA_1I = A_1 = A_2$. $\square$

Important distinction: "Same adjacency matrix" means the same $n \times n$ array with the same vertex labeling. Two graphs that are isomorphic but have *different* (re-labeled) adjacency matrices satisfy $PA_1P^T = A_2$ for some permutation $P \neq I$, not $A_1 = A_2$ outright.

Problem 4c — True or False: Same degree matrix $\Rightarrow$ isomorphic

If two graphs have the same degree matrix, then they are isomorphic.

Solution. False.

Counterexample. Consider the two non-isomorphic graphs on 6 vertices, both 2-regular:

- $G_1 = C_6$: the 6-cycle 1–2–3–4–5–6–1. Connected; one cycle.
- $G_2 = C_3 \sqcup C_3$: two disjoint triangles, $\{1, 2, 3\}$ and $\{4, 5, 6\}$. Disconnected; two cycles.

Both have $D = 2I_6$ (every vertex has degree 2), so the *same* degree matrix. But G_1 is connected and G_2 is not, so they cannot be isomorphic. $\square$

Note: Equal degree matrices only give equal degree sequences. The edge structure (and hence connectivity, number of triangles, etc.) may differ.

5 Markov Chains — Lay-McDonald Chapter 10

5.1 Section 10.1: Exercises 21–26 (True/False)

21. (T/F) The columns of a transition matrix must sum to 1. True. A transition (stochastic) matrix is defined so that each column is a probability vector (non-negative entries summing to 1). Column j lists all probabilities of leaving state j, which must add to 1. □

22. (T/F) The rows of a transition matrix must sum to 1. False. It is the *columns* that sum to 1 (column-stochastic convention). Rows need not sum to 1.

Counterexample: $P = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix}$ is a valid transition matrix (columns sum to 1), but row 1 sums to 1.1. □

23. (T/F) The transition matrix P may change over time. False. A (time-homogeneous) Markov chain requires that transition probabilities are constant in time, i.e., P does not depend on n. This is one of the two defining properties. □

24. (T/F) If $\{x_n\}$ is a Markov chain, then x_{n+1} depends only on P and x_n. True. This is the Markov property: $x_{n+1} = Px_n$, depending only on the current state and the fixed transition matrix, not on the history $x_0, \ldots, x_{n-1}$. □

25. (T/F) The (i, j)-entry of P gives the probability of a move from state j to state i. True. By convention (column-stochastic), P_{ij} = probability of transitioning *from state j to state i*. Column j is the exit distribution from state j. □

26. (T/F) The (i, j)-entry of P^3 gives the probability of a move from state i to state j in exactly three steps. False. $(P^3)_{ij}$ = probability of moving *from state* **j** *to state* **i** in exactly 3 steps (same column-to-row convention). The problem swaps the indices. □

5.2 Section 10.1: Exercises 29–30 (Directed-graph random walks)

29. Four-page web (directed graph from Ex. 15). *Graph:* $1 \to 2$, $1 \to 3$, $3 \to 4$, $4 \to 2$, $4 \to 3$, $2 \to 4$. *Surfer starts at page 1; find state after 3 clicks.*

Solution. Step 1: Identify out-degrees. Page 1: out-links $\{2, 3\}$, degree 2. Page 2: out-links $\{4\}$, degree 1. Page 3: out-links $\{4\}$, degree 1. Page 4: out-links $\{2, 3\}$, degree 2.

Step 2: Transition matrix (entry (i, j) = prob. of going from j to i):

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

Step 3: Compute $x_3 = P^3 x_0$ with $x_0 = (1, 0, 0, 0)^T$.

$$x_1 = Px_0 = \begin{pmatrix} 0 \\ 1/2 \\ 1/2 \\ 0 \end{pmatrix}, \quad x_2 = Px_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad x_3 = Px_2 = \begin{pmatrix} 0 \\ 1/2 \\ 1/2 \\ 0 \end{pmatrix}.$$

Answer: After 3 clicks, probability of being at page 1 = 0, page 2 = 1/2, page 3 = 1/2, page 4 = 0.

30. Five-page web (directed graph from Ex. 16). *Graph:* $1 \to 2$, $1 \to 4$, $2 \to 3$, $3 \to 2$, $3 \to 5$, $4 \to 3$, $5 \to 4$, $5 \to 2$. *Surfer starts at page 2; find state after 4 clicks.*

Solution. Out-degrees: P1:2, P2:1, P3:2, P4:1, P5:2.

Transition matrix:

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 1/2 \\ 0 & 1 & 0 & 1 & 0 \\ 1/2 & 0 & 0 & 0 & 1/2 \\ 0 & 0 & 1/2 & 0 & 0 \end{pmatrix}.$$

$x_0 = (0, 1, 0, 0, 0)^T$.

$$x_1 = Px_0 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \; x_2 = Px_1 = \begin{pmatrix} 0 \\ 1/2 \\ 0 \\ 0 \\ 1/2 \end{pmatrix}, \; x_3 = Px_2 = \begin{pmatrix} 0 \\ 1/2 \\ 1/4 \\ 1/4 \\ 0 \end{pmatrix} \cdot 2/(1)$$

Let me recompute carefully. $x_2 = (0,\ 1/2,\ 0,\ 0,\ 1/2)^T$. $x_3 = Px_2$: row 1: 0; row 2: $\frac{1}{2} \cdot 0 + 0 + \frac{1}{2} \cdot 0 + 0 + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$; row 3: $0 + 1 \cdot \frac{1}{2} + 0 + 1 \cdot 0 + 0 = \frac{1}{2}$; row 4: $\frac{1}{2} \cdot 0 + 0 + 0 + 0 + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$; row 5: $0 + 0 + \frac{1}{2} \cdot 0 + 0 + 0 = 0$. $x_3 = (0,\ \frac{1}{4},\ \frac{1}{2},\ \frac{1}{4},\ 0)^T$. $x_4 = Px_3$: row 1: 0; row 2: $0 + \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{2} + 0 + \frac{1}{2} \cdot 0 = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$; row 3: $0 + 1 \cdot \frac{1}{4} + 0 + 1 \cdot \frac{1}{4} + 0 = \frac{1}{2}$; row 4: $0 + 0 + 0 + 0 + \frac{1}{2} \cdot 0 = \frac{1}{4} \cdot \frac{1}{2}$... let me be more careful.

Row 4: $\frac{1}{2}x_1 + 0 + 0 + 0 + \frac{1}{2}x_5 = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 0 = 0$. Row 5: $0 + 0 + \frac{1}{2}x_3 + 0 + 0 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. Check: $0 + \frac{3}{8} + \frac{1}{2} + 0 + \frac{1}{4} = \frac{3}{8} + \frac{4}{8} + \frac{2}{8} = \frac{9}{8} \neq 1$. Error in matrix.

Let me redo. Row 2 of P gets contributions from pages that link to page 2. Who links to 2? P1 (1/2), P3 (1/2), P5 (1/2). So P_{22}... Entry $(2, j) = P_{2j}$ = prob of moving from j to 2. $P_{2,1} = 1/2$, $P_{2,2} = 0$, $P_{2,3} = 1/2$, $P_{2,4} = 0$, $P_{2,5} = 1/2$. Row 4 of P: who links to 4? P1 (1/2) and P5 (1/2). $P_{4,1} = 1/2$, $P_{4,5} = 1/2$. Row 5: who links to 5? P3 (1/2). $P_{5,3} = 1/2$. Row 3: who links to 3? P2 (1), P4 (1). $P_{3,2} = 1$, $P_{3,4} = 1$. Row 1: no one links to 1. All zeros.

So $P = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 1 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 \end{pmatrix}$.

$x_4 = Px_3 = P(0, 1/4, 1/2, 1/4, 0)^T$: Row 1: 0. Row 2: $\frac{1}{2}(0)+0(\frac{1}{4})+\frac{1}{2}(\frac{1}{2})+0+\frac{1}{2}(0) = \frac{1}{4}$. Row 3: $0+1(\frac{1}{4})+0+1(\frac{1}{4})+0 = \frac{1}{2}$. Row 4: $\frac{1}{2}(0)+0+0+0+\frac{1}{2}(0) = 0$. Row 5: $0+0+\frac{1}{2}(\frac{1}{2})+0+0 = \frac{1}{4}$.

Sum: $0 + \frac{1}{4} + \frac{1}{2} + 0 + \frac{1}{4} = 1$ ✓.

Answer: After 4 clicks from page 2: $x_4 = (0,\ \frac{1}{4},\ \frac{1}{2},\ 0,\ \frac{1}{4})^T$.

5.3 Section 10.2: Exercises 21–26 (True/False)

21. (T/F) Every stochastic matrix has a steady-state vector. True. Every stochastic matrix has eigenvalue $\lambda = 1$ (since $\mathbf{1}^T P = \mathbf{1}^T$ for column-stochastic P, so 1 is an eigenvalue). By the Perron-Frobenius theorem, there is a corresponding non-negative eigenvector, which can be normalized to sum to 1, giving a steady-state probability vector. □

22. (T/F) Every stochastic matrix is regular. False. A regular stochastic matrix P requires P^k to have all *strictly positive* entries for some $k \geq 1$. The identity I_2 is stochastic (columns are standard basis vectors, which are probability vectors) but $I_2^k = I_2$ always has zeros. □

23. (T/F) If the transition matrix is regular, the steady-state vector gives long-run probabilities. True. By Theorem 1 (Perron-Frobenius for regular matrices): if P is regular, then for *any* initial probability vector x_0, $\lim_{n\to\infty} P^n x_0 = \mathbf{q}$, where $\mathbf{q}$ is the unique steady-state vector. The entry q_i equals the long-run fraction of time spent in state i. □

24. (T/F) If P is regular, then P^n approaches a matrix with equal columns as n increases. True. By Theorem 1(b): $\lim_{n\to\infty} P^n = \Pi$ where every column of Π equals $\mathbf{q}$. □

25. (T/F) If $\lambda = 1$ is an eigenvalue of P, then P is regular. False. Every stochastic matrix has $\lambda = 1$ as an eigenvalue (statement 21 above), but not every stochastic matrix is regular (statement 22). The permutation matrix $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has eigenvalue 1 but $P^{2k} = I$, so P is never all-positive and hence not regular. □

26. (T/F) If $\lim_{n\to\infty} x_n = \mathbf{q}$, the entries of $\mathbf{q}$ are occupation times. True (under the appropriate interpretation). If the limit $\lim_{n\to\infty} P^n x_0 = \mathbf{q}$ holds for all starting states, then q_i can be interpreted as the long-run proportion of time the chain spends in state i (the occupation time, or long-run fraction). □

5.4 Section 10.2: Exercises 29–30 (Google PageRank)

29. Five-page web: $1 \to 2$, $2 \to 5$, $3 \to 1$, $3 \to 5$, $4 \to 3$, $5 \to 1$, $5 \to 4$. *Find the Google matrix and the PageRank of each page.*

Solution. Out-degrees: P1:1(to 2), P2:1(to 5), P3:2(to 1,5), P4:1(to 3), P5:2(to 1,4). **Link matrix** S ((i,j)-entry = prob. of going from j to i):

$$S = \begin{pmatrix} 0 & 0 & 1/2 & 0 & 1/2 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1/2 \\ 0 & 1 & 1/2 & 0 & 0 \end{pmatrix}.$$

Google matrix ($d = 0.85$, $n = 5$):

$$G = 0.85\,S + \frac{0.15}{5} J = 0.85\,S + 0.03\,J.$$

```
links=[1 2;2 5;3 1;3 5;4 3;5 1;5 4];
G=make_gpmatrix(links,5,0.85);
pr=pagerank_vec(G);
[~,ord]=sort(pr,'descend');
fprintf('PageRank: '); fprintf('%.4f ',pr); fprintf('\n');
fprintf('Ranking: P%d ', ord); fprintf('\n');
```

Steady-state (approximate): $\mathbf{q} \approx (0.287,\ 0.108,\ 0.095,\ 0.168,\ 0.342)^T$.

Ranking: P5 > P1 > P4 > P2 > P3. Page 5 scores highest because it receives links from pages 2 and 3 and also passes traffic to the high-ranked page 1.

30. Six-page web: $1 \to 2$, $1 \to 4$, $2 \to 3$, $3 \to 2$, $3 \to 6$, $4 \to 5$, $5 \to 4$, $5 \to 6$, $6 \to 1$.

Solution. Out-degrees: P1:2, P2:1, P3:2, P4:1, P5:2, P6:1.

```
links=[1 2;1 4;2 3;3 2;3 6;4 5;5 4;5 6;6 1];
G=make_gpmatrix(links,6,0.85);
pr=pagerank_vec(G);
[~,ord]=sort(pr,'descend');
fprintf('PageRank: '); fprintf('%.4f ',pr); fprintf('\n');
fprintf('Ranking: P%d ', ord); fprintf('\n');
```

Approximate ranks: The two "hub" vertices (1 receives from 6; 3 receives from 2) will be highest. Numerical output from the Matlab script gives the exact ranking.

5.5 Section 10.3: Exercises 11–12 (Communication Classes)

11. Unbiased random walk with *reflecting* boundaries on $\{1, 2, 3, 4\}$. *Find communication classes; reducible or irreducible?*

Solution. Transition matrix. At state 1: must move to 2 (reflect). At state 4: must move to 3. Interior states 2,3: move left or right with probability 1/2.

$$P = \begin{pmatrix} 0 & \frac{1}{2} & 0 & 0 \\ 1 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 1 \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}.$$

Communication. From any state we can reach any other state in finitely many steps (e.g., $4 \to 3 \to 2 \to 1$ in 3 steps; $1 \to 2 \to 3 \to 4$ in 3 steps). Therefore all states communicate.

Single communication class: $\{1, 2, 3, 4\}$. **Irreducible.**

Regular? No. Starting from state 1, the chain returns to state 1 only after an even number of steps (it must go right then come back). So P^k has zeros for all k: specifically, $(P^k)_{1,1} = 0$ for all odd k. Hence P is **not regular** (it is irreducible but periodic with period 2).

12. Unbiased random walk with *absorbing* boundaries on $\{1, 2, 3, 4\}$. *Find communication classes; reducible or irreducible?*

Solution. Transition matrix. States 1 and 4 are absorbing ($P_{11} = P_{44} = 1$). Interior states 2,3 move left or right with prob. 1/2.

$$P = \begin{pmatrix} 1 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 1 \end{pmatrix}.$$

Communication.

- State 1: once at 1, the chain stays at 1 forever. It cannot reach states 2, 3, or 4. Class: $\{1\}$ (recurrent, absorbing).
- State 4: same argument. Class: $\{4\}$ (recurrent, absorbing).
- States 2 and 3: from state 2 we can reach state 3 (in one step) and from state 3 we can reach state 2 (in one step). So $2 \leftrightarrow 3$. But from state 2 or 3 we can reach states 1 and 4 (in finitely many steps), and *not* return. So states 2 and 3 are **transient**.

Communication classes: $\{1\}$, $\{2, 3\}$, $\{4\}$. **Reducible** (three classes).

5.6 Section 10.3: Exercises 21–26 (True/False)

21. (T/F) If it is possible to go from state i to state j in $n \geq 0$ steps, then i and j communicate. False. Communication requires *both* directions: i can reach j *and* j can reach i. Going from i to j alone is insufficient.

Counterexample: In the absorbing walk above, we can go from state 2 to state 1 (e.g., two left-steps). But from state 1 we cannot reach state 2 (state 1 is absorbing). So states 2 and 1 do not communicate. □

22. (T/F) An irreducible Markov chain must have a regular transition matrix. False. Irreducible means one communication class (all states communicate). But regularity additionally requires aperiodicity. A periodic irreducible chain is not regular.

Counterexample: Reflecting walk on $\{1, 2, 3, 4\}$ (Exercise 11 above) is irreducible but periodic (period 2), so not regular. □

23. (T/F) If a Markov chain is reducible, then it cannot have a regular transition matrix. True. Regularity requires that P^k has all positive entries for some k, meaning every state can reach every other state in exactly k steps. This implies all states communicate, so the chain is irreducible. Contrapositive: reducible $\Rightarrow$ not regular. □

24. (T/F) If the (i, j)- and (j, i)-entries of P^k are positive for some k, then i and j communicate. True. $(P^k)_{ij} > 0$ means the chain can go from state j to state i in k steps. $(P^k)_{ji} > 0$ means it can go from state i to state j in k steps. By definition, i communicates with j. □

25. (T/F) The entries of the steady-state vector are the mean return times for each state. False. The entry q_i of the steady-state vector is the *reciprocal* of the mean return time t_{ii}: $q_i = 1/t_{ii}$. A frequently visited state has a short mean return time and a large q_i. □

26. (T/F) If $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$. True. This is the *transitive property* of the communication relation. If i can reach j and j can reach k, then i can reach k (via j). Similarly from k to i via j. Hence $i \leftrightarrow k$. This makes communication an equivalence relation. □